# BALWINDER HAYER

+1-949-297-6501 | BSHAYER@LIVE.COM | Lake Forest, CA

## OBJECTIVE

Engineering Technician and QA professional with hands-on experience in firmware/hardware validation, failure analysis, and automated test development for HDD and embedded systems, plus a background in software QA and process validation in regulated manufacturing. Skilled in Python, C++, and TCL scripting, hardware debug instrumentation, and root-cause problem solving, with a track record of building test automation and collaborating across engineering teams.

## EXPERIENCE

### Western Digital Corporation — Irvine, CA — August 2021 – 2026

*ADVE Engineering*

- Supported ADVE with Complete firmware update procedure using Niagara in-built tcl/python updating binaries, automated debug collections for ETL/memdump, after error logs, logged to JIRA, debug, perform root-cause analysis approach.
- Support firmware and hardware validation and debug for ARCH6, CCB hard drives.
- Perform failure analysis based on ShowStop and debug snapshots, creating JIRA epics to track issue, debug, and root-cause issues; collect bus traces with a serial debugger, signal analyzer, and logic analyzer as needed.
- Triage system-level HDD failures using blink codes, error logs, and firmware Event Trace Logging (ETL), applying system-level knowledge of firmware and hardware to isolate root cause by design area.
- Debug hardware and PCBA designs using oscilloscopes, signal generators, logic/bus analyzers, embedded trace, test muxes, and GHS Probe; review schematics for high-speed signal integrity issues.
- Run/review hard drive performance SIT/Windows based testing on equivalent pairs of hard drives, provide benchmark results to DevOps
- Perform hard drive performance testing based on TRUNK, Customer Release, OEM, custom based customer tests i.e. WASABI, AWS, MSFT specific high capacity HAMR drives
- Developed and maintained an automated firmware/hardware test validation suite (TMC/SIT) and Windows applications for ADVE performance testing, writing reusable C++, TCL, and Python scripts and CLI tools for builds, testing, and design-of-experiments. i.e. ATTO, CDM5, PCM Suites, IOMeter compatibility tools in native powershell environment.
- Set up Jenkins/Ubuntu pipelines for unattended overnight hardware test runs, apply A3/8D problem-solving, and collaborate across engineering teams, geographies, and cultures to analyze complex issues and deliver clear, concise solutions.

### B. Braun Medical Inc. (P&G) — Irvine, CA — 2018 – 2021

*Quality Assurance Analyst*

- Developed and maintained automated test cases ensuring comprehensive testing of web-based and desktop software.
- Worked with test management tools (Zephyr, TestLink) and defect/task management tools (Jira, Bugzilla).
- Performed test case design, test data preparation, XML validation, and proper storage in QMETRY.
- Analyzed business requirements and functional/technical designs to develop test strategies, test plans, scenarios, test cases, traceability matrices, and test analysis reports.
- Installed and configured recreations of software production environments to test software performance.
- Validated Sterilizer Load Reports (SLR) based on temperature mapping and validated data in EBR.
- Worked with SQL Server to extract data for manual batch release using Historian.
- Conducted timely review of SLR report data generated by SCADA/PLC systems to support batch release.
- Performed initial debugging by reviewing configuration files, logs, and code to determine the source of issues.

## EDUCATION

### Bachelor of Science in Computer Science — 2020

Coursework in software development, data structures, systems programming, and computer architecture — foundation for firmware/hardware debug, automation scripting, and test tool development in later roles.

### Associate of Arts, Emphasis in Liberal Studies — Irvine Valley College, Irvine, CA — 2019

Communications, analytical writing, and social sciences, supporting the collaboration and clear communication used in cross-functional engineering and QA work.

## SKILLS

**Programming & Scripting:** C++, Python, TCL

**Firmware/Hardware Debug:** Error logs, ShowStop/debug snapshots, bus traces, Event Trace Logging (ETL), ETL/memdump collection via Niagara

**Test & Debug Instrumentation:** Oscilloscopes, signal generators, logic analyzers, bus analyzers, embedded trace, GHS Probe, serial debuggers

**Hardware & Systems Knowledge:** High-speed PCB signal integrity, schematic review, system-level HDD firmware/hardware architecture

**Test Automation & CI/CD:** TMC/SIT test suites, Jenkins, GitLab, Automated Regression, SIT/WinOS Performance testing

**QA Tools & Processes:** Jira, Bugzilla, Zephyr, TestLink, QMETRY, test case design, traceability matrices, test plans and analysis reports

**Problem Solving & Methodologies:** A3, 8D, Root-cause Analysis, design-of-experiments (DOE)

**Collaboration:** cross-functional and cross-geography teamwork, clear technical communication, project management under tight deadlines